



— DOCUMENT SEARCH

The most popular retrieval methods for PDF RAG

BM25 · Vector · Text Hybrid · Multimodal Hybrid ·
Multivector Image · MUSERA

01

Keyword

BM25 SEARCH

Scores **keywords** (exact matches) based on how often word appears in a document weighted against how rare the term is in the corpus

query

transformers **model**

score

DOCUMENT

 = 4 = 5

X

CORPUS

 = 159 = 2630

document score = 0.86

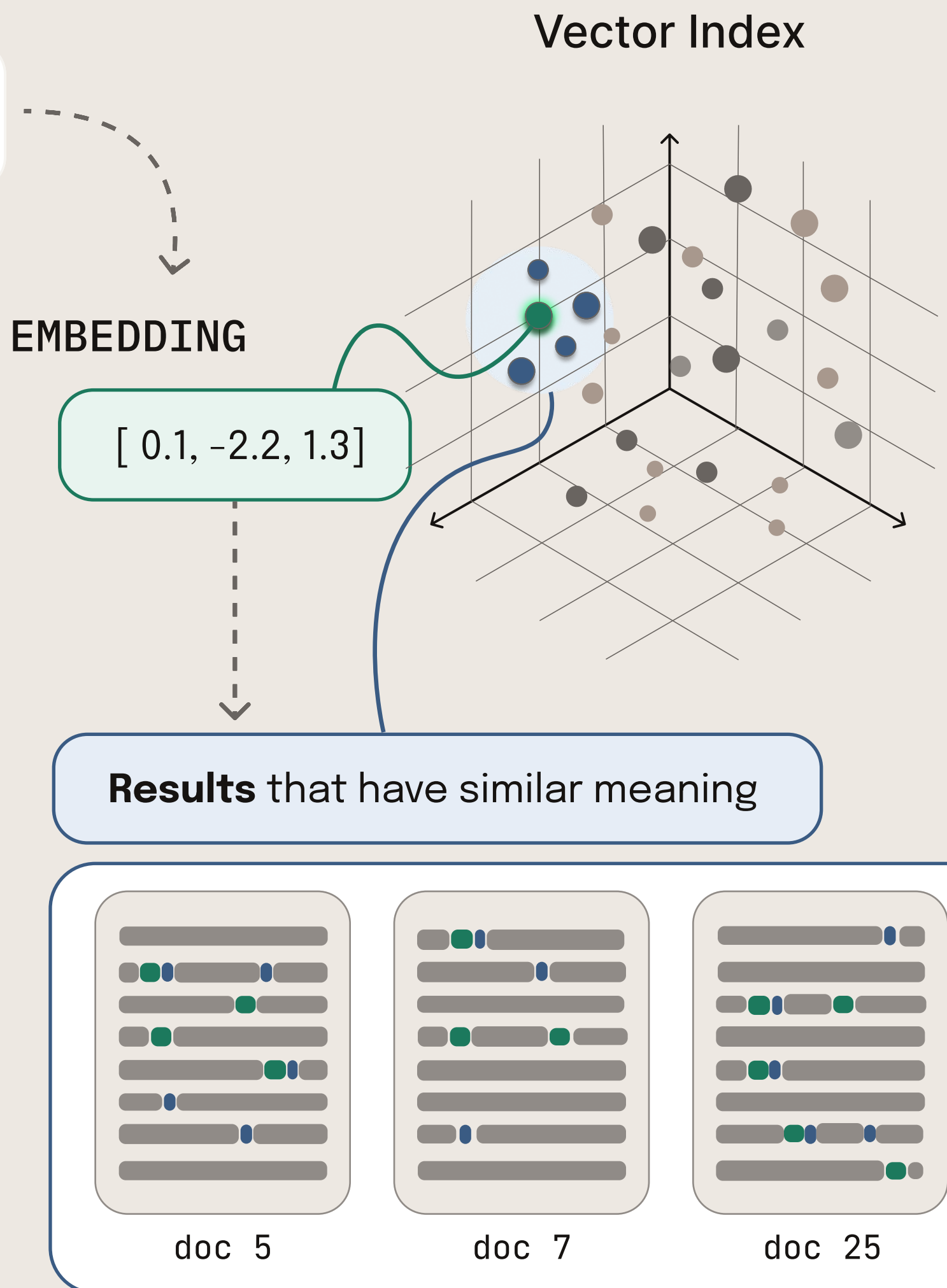
02

Vector

DENSE VECTOR EMBEDDINGS

query

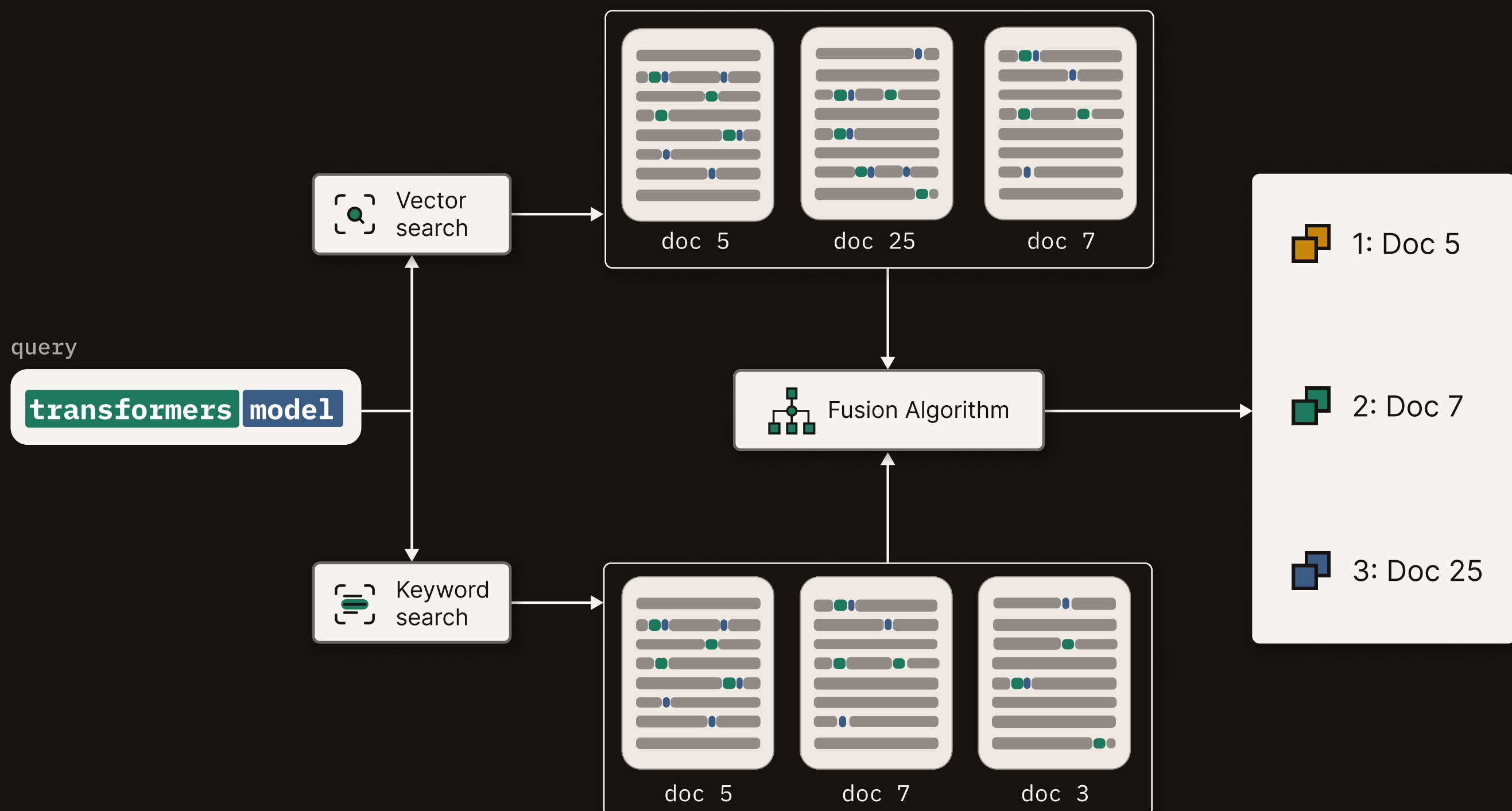
transformers model



Scores documents based on how **semantically related** they are to the query, calculated through vector embedding distance.

Hybrid text

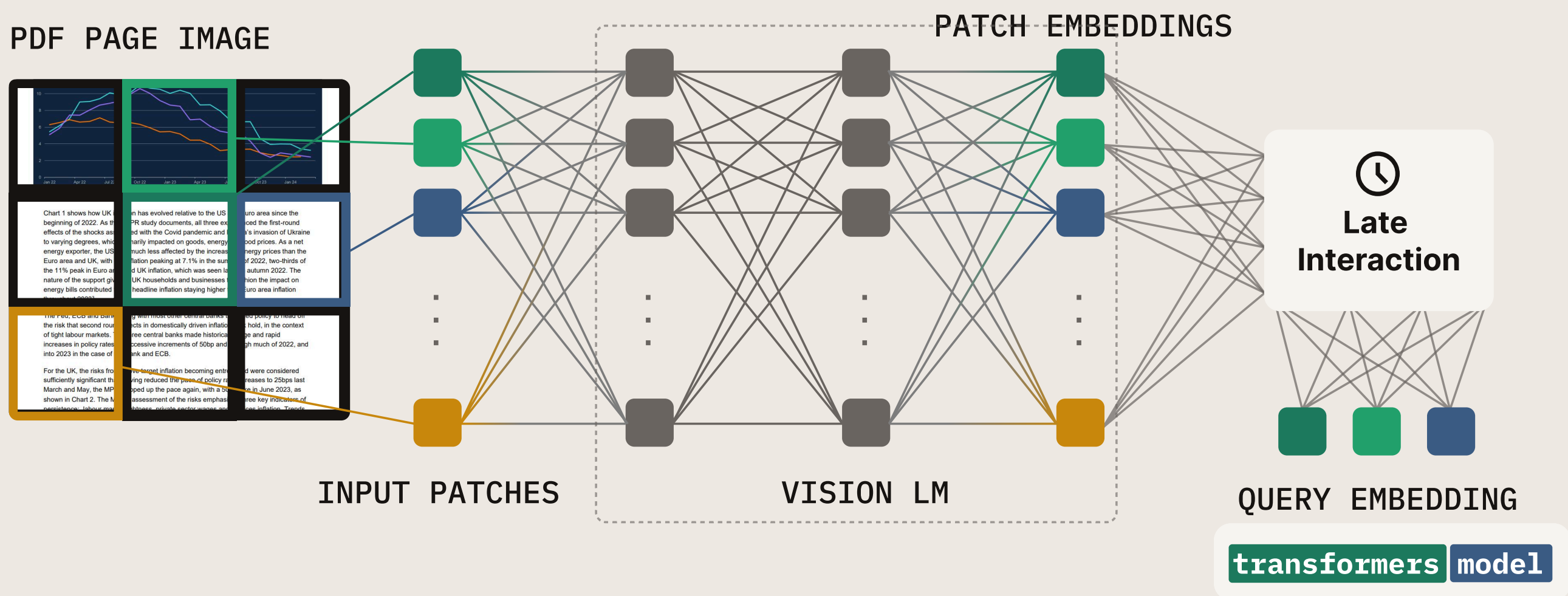
KEYWORD + VECTOR SEARCH



Computes **both vector search and keyword search in parallel**, then combines both scores in a total document score to rerank accordingly.

Multivector Image

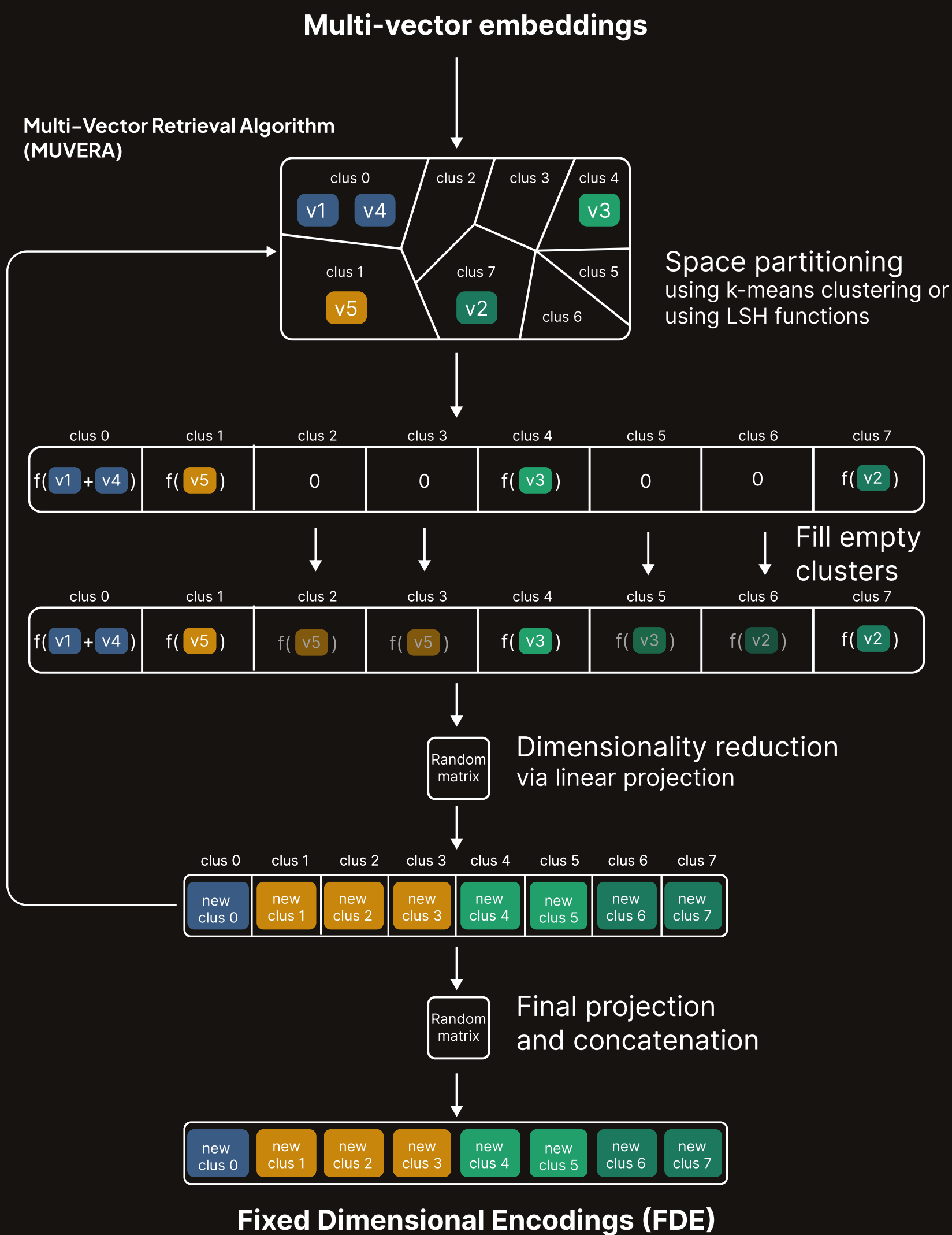
COLMODERNVBERT, COLPALI, COLQWEN



Segments **page images into patches**, then embeds each patch. Patch embeddings are scored against query token embeddings using late interaction to compute the document score.

Compressed Multivector

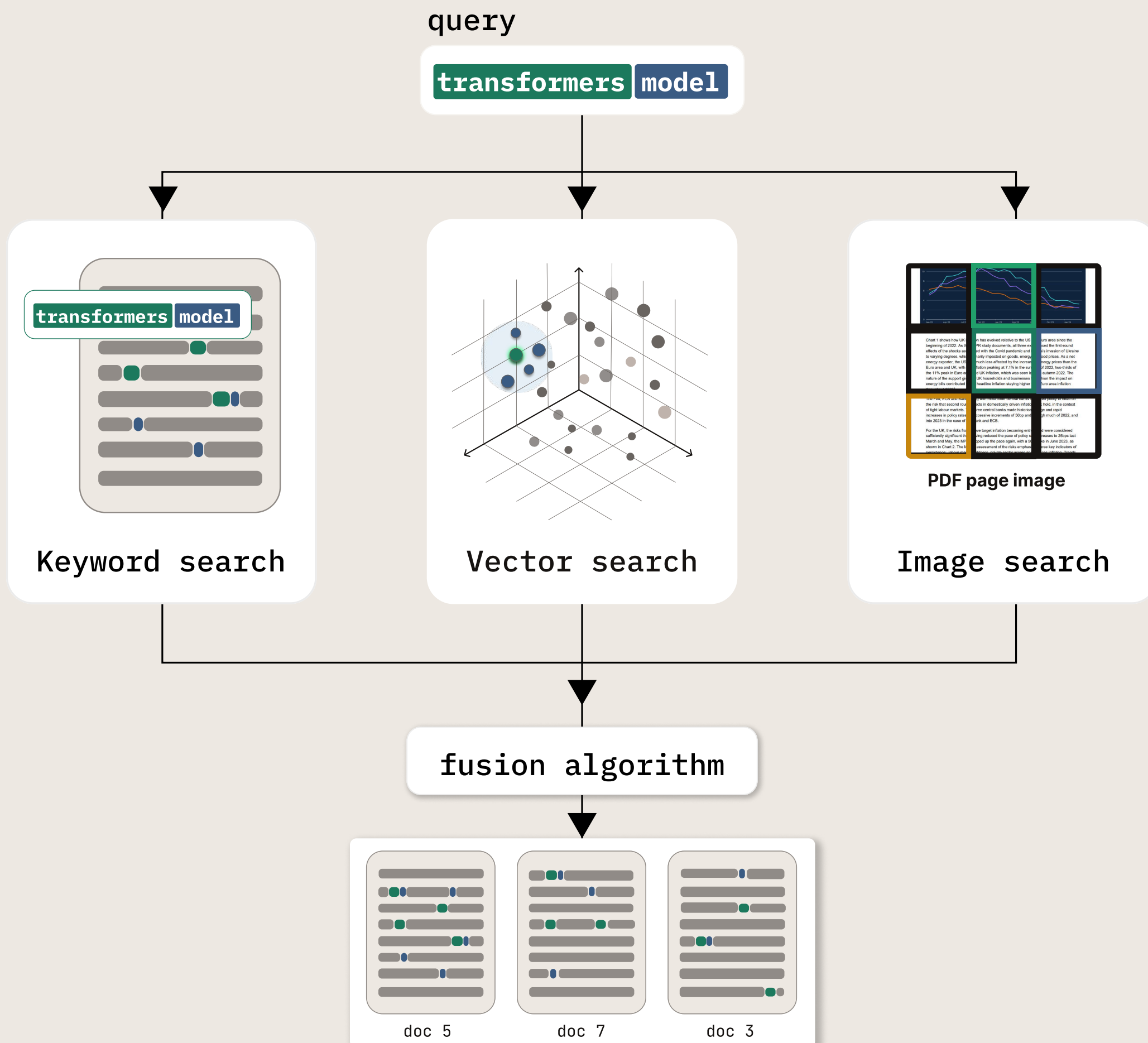
MUVERA + MULTIVECTOR EMBEDDING MODELS



Instead of storing
one vector
embedding per
image patch/
token, MUVERA
compresses
multivector
embeddings into a
**single document
embedding (FDE)**
through a
clustering
algorithm.

Hybrid Multimodal

KEYWORD + VECTOR + IMAGE SEARCH



Computes **keyword, vector and image document scores** in parallel, then combines them using a fusion algorithm.

Check out this video to see how they compare.

IRPAPERS: A VISUAL DOCUMENT BENCHMARK FOR SCIENTIFIC RETRIEVAL AND QUESTION ANSWERING

